

Artificial intelligence bike that self-balances, avoids obstacles and responds to voice commands

Chinese scientists have developed a ground-breaking electronic chip and incorporated it into a rider-less autonomous bicycle. The bicycle can detect and track targets, avoid obstacles, self-balance, understand voice commands and even make independent decisions as a result of the chip's simultaneous processing of versatile algorithms and models.



Tianjic chip shows a hybrid architecture capable of supporting so-called artificial general intelligence (Credit: Tsinghua University)

There are two main approaches to developing artificial general intelligence. One is rooted in neuroscience and attempts to construct circuits that closely mimic the brain. The other is grounded in computer science and uses computers to execute machine learning algorithms.

Tianjic chip can integrate the two approaches into one hybrid platform, which enables it to accommodate both machine learning algorithms and brain-inspired circuits.

This research was conducted by Shi Luping and his colleagues at Center for Brain Inspired Computing Research of Tsinghua University.

After continuous improvement, the current second-generation Tianjic chip features even higher performance with much lower power consumption. "Compared with similar chips in the market, our chip's density is twenty per cent higher, (it is) at least ten times faster and its bandwidth has been increased by at least hundred times," explains Luping.

He adds, "Tianjic chip aims to provide a more efficient, fast and flexible computing platform for the research of artificial general intelligence, as well as the development of various applications."

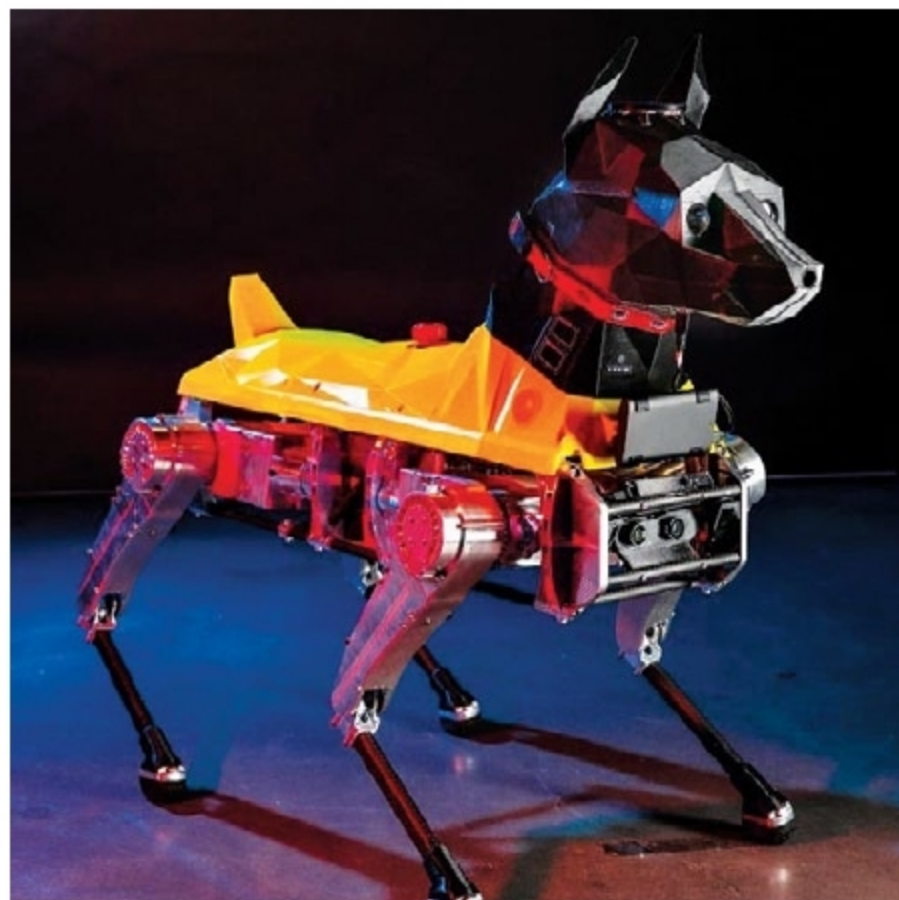
Astro is a four-legged seeing and hearing intelligent robodog

Using deep learning and artificial intelligence (AI), scientists from Florida Atlantic University's (FAU's) Machine Perception and Cognitive Robotics Laboratory (MPCR) in Center for Complex Systems and Brain Sciences in FAU's Charles E. Schmidt College of Science are developing a handful of quadruped robots. Astro, the smart robodog, is unique because he is the only one of these robots with a head, 3D-printed to resemble a Doberman pinscher, that contains a (computerised) brain.

Astro not only looks like a dog, he learns like one, too. That is because he does not operate based on pre-programmed robotic automation. Instead, Astro is being trained using inputs to a deep neural network—a computerised simulation of a brain—so that he can learn from experience to perform human-like tasks, or, in his case, dog-like tasks, that benefit humanity.

Equipped with sensors, high-tech radar imaging, cameras and a directional microphone, this 45.4kg (100-pound) super robot is still a puppy-in-training. Just like a regular dog, he responds to commands such as sit, stand and lie down. Eventually, he will be able to understand and respond to hand signals, detect different colours, comprehend many languages, coordinate his efforts with drones, distinguish human faces and even recognise other dogs.

As an information scout, Astro's key missions will include detecting guns, explosives and gun residue to assist police, the military and security personnel. He also can be programmed to assist as a service dog for the visually-impaired, or to provide medical diagnostic monitoring. Scientists are also training



Designed to engage and react to the world around him in real time, Astro will be able to navigate through rough terrains and respond to dangerous situations to keep humans and animals out of harm's way (Credit: Florida Atlantic University)

Astro to serve as a first responder for search-and-rescue missions such as hurricane reconnaissance as well as military manoeuvres.